

THE POLYGONAL BIGRAPHS $P_{8s,4s-2}$ ARE 2-EDGE-HAMILTONIAN

ABSTRACT. Simmons conjectured that every polygonal bigraph $P_{m,k}$ is 2-edge-Hamiltonian. We prove the conjecture for the infinite family $P_{8s,4s-2}$, $s \geq 1$, by an explicit four-edge switch and cyclic rotations; the cases not already covered by Simmons are exactly $s \equiv 1 \pmod{3}$ with $s \geq 4$. For those parameters the graphs lie in the arithmetic residue $2 < \gcd(m+1, k+1) < k+1$ isolated in [1, p. 109]. The first such case is $P_{32,14}$; the table of the addendum [2] leaves that cell blank, so our theorem supplies the one case needed for that addendum's assertion that all $P_{m,k}$ on 66 or fewer vertices are 2-edge-Hamiltonian. The remaining parameters are not addressed.

1. THE CONSTRUCTION

A graph is *2-edge-Hamiltonian* if every pair of distinct edges lies on a common Hamilton cycle. Simmons defined $P_{m,k}$ from a cycle on $2m$ vertices by joining each odd vertex, in a one-based clockwise labeling, to the even vertex $2k+1$ positions ahead [1, pp. 101–102]. He conjectured that every $P_{m,k}$ is 2-edge-Hamiltonian [1, p. 101]. The map $v \mapsto -v \pmod{2m}$ is an isomorphism between Simmons's orientation and the one used below, in which each even vertex is joined to the odd vertex $2k+1$ positions ahead, so we may work with the latter.

Theorem 1. *For every integer $s \geq 1$, the polygonal bigraph $P_{8s,4s-2}$ is 2-edge-Hamiltonian.*

Proof. Put

$$m = 8s, \quad k = 4s - 2 = \frac{m}{2} - 2.$$

Work modulo $2m$ on vertices and modulo m on subscripts. The three natural perfect matchings of $P_{m,k}$ are

$$A_i = \{2i, 2i+1\}, \quad B_i = \{2i+1, 2i+2\}, \quad C_i = \{2i, 2i+2k+1\}.$$

Write $A = \{A_i : i \in \mathbb{Z}/m\mathbb{Z}\}$, and similarly for B and C .

The outer cycle $A \cup B$ is Hamilton. The subgraph $B \cup C$ is also Hamilton, because a chord followed by a B -edge sends the even subscript

$$i \mapsto i + k + 1,$$

and

$$\gcd(k+1, m) = \gcd(4s-1, 8s) = 1.$$

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These two Hamilton cycles cover every edge pair except pairs of the form $\{C_p, A_q\}$.

A chord followed by an A -edge sends the even subscript i to $i + k$. Since

$$\gcd(k, m) = \gcd(4s - 2, 8s) = 2,$$

the subgraph $A \cup C$ consists of two cycles, one on the even subscripts and one on the odd subscripts. Define

$$H = (A \setminus \{A_k, A_{k+m/2}\}) \cup (C \setminus \{C_{k+1}, C_{k+1+m/2}\}) \\ \cup \{B_{k-1}, B_k, B_{k-1+m/2}, B_{k+m/2}\}.$$

We show that H is a Hamilton cycle. For $\varepsilon \in \{0, 1\}$, put $i_t = \varepsilon + tk$ in $\mathbb{Z}/m\mathbb{Z}$. The corresponding component of $A \cup C$ has cyclic order

$$(2i_0, 2i_1 + 1, 2i_1, 2i_2 + 1, \dots, 2i_{m/2-1}, 2i_0 + 1).$$

Since

$$\frac{m}{4}k \equiv \frac{m}{2} \pmod{m},$$

the two deleted edges in each component occur halfway around that component. Reading the two cyclic orders gives four spanning paths with endpoint pairs

$$\{2k + 1, 2k + m\}, \quad \{2k, 2k + m + 1\}, \\ \{2k + 2, 2k - 1\}, \quad \{2k + m - 1, 2k + m + 2\}.$$

Here the second line uses $k = m/2 - 2$, so

$$4k + 3 = 2k + m - 1, \quad 4k + m + 3 \equiv 2k - 1 \pmod{2m}.$$

The four added edges join the paths in the cyclic order

$$2k + 1 \xrightarrow{B_k} 2k + 2 \rightsquigarrow 2k - 1 \xrightarrow{B_{k-1}} 2k \rightsquigarrow 2k + m + 1 \\ \xrightarrow{B_{k+m/2}} 2k + m + 2 \rightsquigarrow 2k + m - 1 \xrightarrow{B_{k-1+m/2}} 2k + m \rightsquigarrow 2k + 1.$$

Thus H is spanning, connected, and 2-regular, hence Hamiltonian.

Now let C_p and A_q be arbitrary and put $d = q - p \in \mathbb{Z}/m\mathbb{Z}$. Choose

$$i \notin \{k + 1, k + 1 + m/2, k - d, k + m/2 - d\}.$$

Such an i exists because $m \geq 8$. Then H contains both C_i and A_{i+d} . The rotation $\rho(v) = v + 2$ is an automorphism satisfying

$$\rho^r(C_i) = C_{i+r}, \quad \rho^r(A_j) = A_{j+r}.$$

Taking $r = p - i$, the Hamilton cycle $\rho^r(H)$ contains C_p and $A_{p+d} = A_q$. Hence every remaining edge pair lies on a Hamilton cycle. $\square$

Corollary 2. *For every integer $t \geq 1$, the graph $P_{24t+8,12t+2}$ is 2-edge-Hamiltonian and satisfies*

$$2 < \gcd(m + 1, k + 1) < k + 1.$$

In particular, the first member is $P_{32,14}$.

Proof. Set $s = 3t + 1$ in Theorem 1. Then

$$m = 24t + 8, \quad k = 12t + 2, \quad \gcd(m + 1, k + 1) = 3.$$

Since $k + 1 = 12t + 3 > 3$, the strict inequalities follow. This is precisely the residue $2 < \gcd(m + 1, k + 1) < k + 1$ left open in [1, p. 109] and re-examined in [2, p. 102]. $\square$

Proposition 3. *For every $s \geq 1$, the graph $P_{8s,4s-2}$ is not isomorphic to $P_{8s,k'}$ for any $k' \neq 4s - 2$ in the canonical range $1 \leq k' \leq 4s - 1$.*

Proof. For fixed m , the bipartite adjacency block of $P_{m,k}$ is a weight-three circulant with support affinely equivalent to

$$S_k = \{0, 1, k + 1\} \subseteq \mathbb{Z}/m\mathbb{Z}.$$

An isomorphism between two connected bipartite graphs either preserves or interchanges their parts; interchanging the parts transposes the block and replaces its support by its negative. It follows from [4, Theorem 1.1] that isomorphic polygonal bigraphs have affinely equivalent supports.

For the family above, put

$$a = k + 1 = \frac{m}{2} - 1.$$

Then a is a unit modulo m , $a - 1$ is not, and

$$a^2 \equiv 1 \pmod{m}.$$

Consequently, after normalizing an ordered pair at unit distance to 0, 1, the third element is one of

$$a, \quad 1 - a, \quad a^{-1}, \quad 1 - a^{-1}.$$

Since $a^{-1} = a$, only a and $1 - a$ occur. If $S_{k'} = \{0, 1, b\}$ with $b = k' + 1$ and $2 \leq b \leq m/2$ is affinely equivalent to S_k , then

$$b \in \{a, 1 - a\}.$$

But $1 - a \equiv m/2 + 2 \pmod{m}$, which is outside this range. Hence $b = a$ and $k' = k$. $\square$

The addendum concludes that “all $P_{m,k}$ on 66 or fewer vertices are 2-edge-Hamiltonian” [2, p. 103]. Its table carries no entry at $(m, k) = (32, 14)$, although $2 < \gcd(33, 15) = 3 < 15$; by Proposition 3 the graph $P_{32,14}$ is alone in its isomorphism class, consistently with the count of 9 classes for $m = 32$ in [1, Table 3], and its 64 vertices put it beyond the 56 or fewer vertices reported in [1, p. 115]. Theorem 1 therefore supplies the one case needed for that 66-vertex assertion. The next pair of that residue in our family, $P_{56,26}$, lies beyond the addendum’s range. The companion note [3] in this collection gives an independent covering certificate for $P_{32,14}$ and settles the isomorphism class $P_{34,9} \cong P_{34,14}$, which the addendum records as not known to be 2-edge-Hamiltonian.

REFERENCES

- [1] G. J. Simmons, *Two related (?) 2-edge-Hamiltonian bigraph conjectures*, Bull. Inst. Combin. Appl. **88** (2020), 98–117.
- [2] G. J. Simmons, *Two related (?) 2-edge-Hamiltonian bigraph conjectures addendum*, Bull. Inst. Combin. Appl. **89** (2020), 102–103.
- [3] *Two residual cases of Simmons’s polygonal bigraph conjecture*, companion note in this collection, file **residual-polygonal-bigraphs-close-simmons-finite-range**.
- [4] D. Wiedemann and M. E. Zieve, *Equivalence of sparse circulants: the bipartite Ádám problem*, arXiv:0706.1567, 2007.